The winch model in AWETrim

Force law, drum inertia and drivetrain friction

1 Model

The kite is quasi-steady: at each instant its force balance gives the tether tension T at the ground for a given reel speed v_r. The winch closes the radial direction. Between the tether and the motor sits the drivetrain: drum, gearbox and motor, all referred to the tether. The drum obeys

$$m\,\dot{v}_r = T - F_f(v_r, T) - F_m, \tag{1}$$

with F_m the motor (generator) force, m the rotating inertia seen at the tether and F_f the friction:

$$m = J\left(\frac{n}{r_d}\right)^2, \qquad F_f = (F_c + k\,T)\tanh\!\left(\frac{v_r}{\varepsilon}\right) + c_v\,v_r. \tag{2}$$

J is the total inertia on the motor side, n the gear ratio and r_d the drum radius. F_c is the Coulomb friction, c_v the viscous coefficient and k an optional load-proportional part (pulleys, belt, gear mesh). The tanh, with $\varepsilon = 0.05\,\mathrm{m\,s^{-1}}$, smooths the sign of v_r, so the model stays twice differentiable where v_r changes sign.

The force law sets the motor force. The winch controller is a torque-controlled machine: its law $F_{\mathrm{law}}(v_r, u_p)$ (linear or quadratic in v_r, soft-capped at the force limits, offset by the depower u_p) is the force the motor applies. With Eq. (1), the tension the kite feels is

$$T = F_{\mathrm{law}}(v_r) + F_f(v_r) + m\,\dot{v}_r. \tag{3}$$

Friction raises the tension during reel-out and lowers it during reel-in; inertia adds tension while the drum accelerates. With $m = F_c = c_v = k = 0$ (the default) Eq. (3) is the ideal winch, $T = F_{\mathrm{law}}$.

Shaft power. The power at the motor is

$$P_m = F_m v_r = T v_r - \underbrace{v_r F_f}_{\geq 0} - \frac{\mathrm{d}}{\mathrm{d}t}\left(\tfrac{1}{2} m v_r^2\right). \tag{4}$$

Friction is lost in both reel directions. The inertia term is the drum's kinetic energy: it is returned over a closed cycle, so on a periodic cycle inertia changes the mean power only through the force limits.

2 Discretisation

Forward simulation. The march solves, at each node, the kite's force balance and Eq. (3) for $(T, u_s, \dot{s}, v_r)$. The drum's time constant, m over the slope of the law ($\sim$0.06 s for the LEI-V3), is far shorter than a node step ($\sim$0.3 s), so $\dot{v}_r$ is the backward difference to the previous node (implicit Euler), $\dot{v}_r \approx (v_{r,i} - v_{r,i-1})/\Delta t_{i-1}$. The previous reel speed and $1/\Delta t$ enter the node solver as two parameters; the first node starts with a steady drum.

Optimisation. The cycle NLP uses the same model. Its objective is the mean shaft power, with the inertia part written as the exact change of $\frac{1}{2} m v_r^2$ over each interval, so it telescopes. With the force law (`winch_mode: force_law`) Eq. (3) replaces the law row, with the same backward difference as the march. With a free reel speed (`free_speed`) the motor force $F_m = T - F_f - m\dot{v}_r$ is held below `max_tether_force`.

3 Parameters and use

Table 1: LEI-V3 drivetrain: the TU Delft GS1 drivetrain as V3Kite.jl flies it (`sim_settings_reelout.yaml`, the KiteUtils defaults; Fechner 2016, Table 4.1). Stored tether-referred in `system.yaml`, `drums[0]`.

Quantity	`system.yaml` key	Value
Inertia at the tether, $0.204\,(6.2/0.1615)^2$	`inertia_equivalent_mass`	300.7 kg
Coulomb friction F_c	`friction_coulomb`	122 N
Viscous friction c_v	`friction_viscous`	30.6 N s m^{-1}
Load fraction k (not in V3Kite)	`friction_load_fraction`	0.0535

The drivetrain is off unless `sim_parameters` asks for it: `winch_drivetrain: true` takes m, F_c and c_v from the system file; `winch_inertia_mass`, `winch_friction` and `drivetrain_load_fraction` set the parts one by one (`true` or a value). The cycle script takes `--winch-drivetrain`. Code: `awetrim.system.winch.Drivetrain` and `Winch.radial_equation`.

4 Example

Figure 1 marches the 9 m s^{-1} LEI-V3 cycle with its force law, without optimising. The motor force stays at the law's limit while the tension reaches 8.87 kN, the extra being friction (∼0.23 kN in reel-out) and the drum accelerating into reel-out (up to 0.74 kN). The mean shaft power is 9.8 kW, against 11.0 kW at the tether.

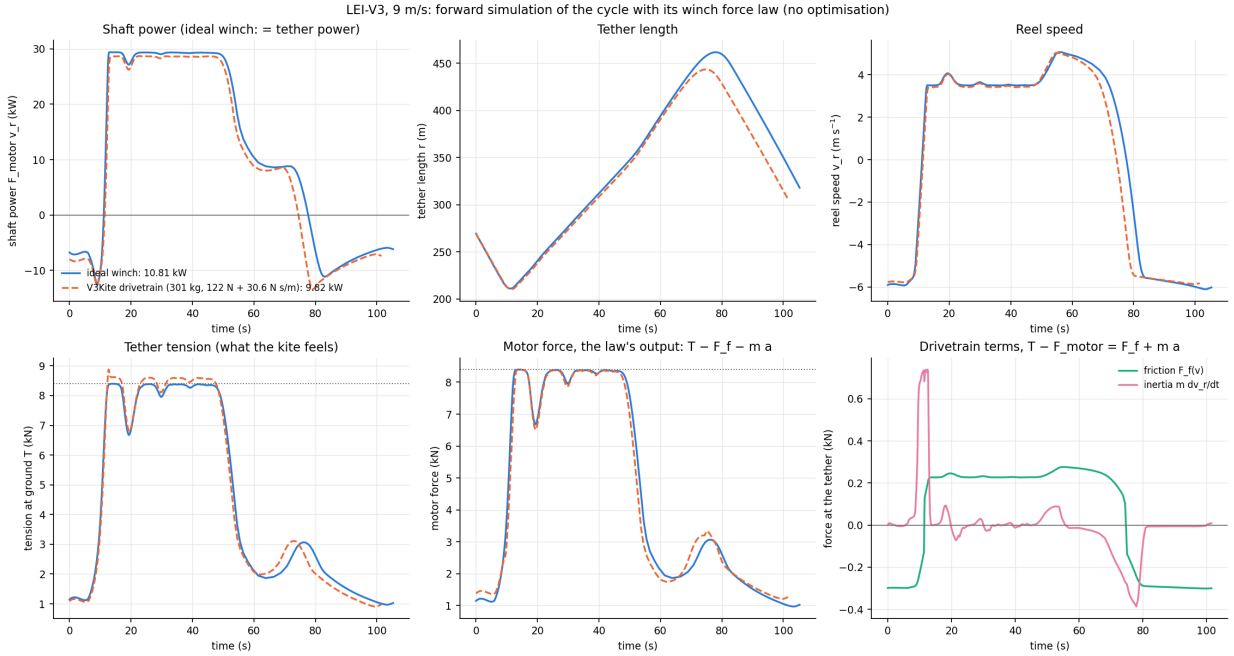


Figure 1: Forward simulation of the 9 m s^{-1} cycle path with an ideal winch and with the LEI-V3 drivetrain (Table 1).